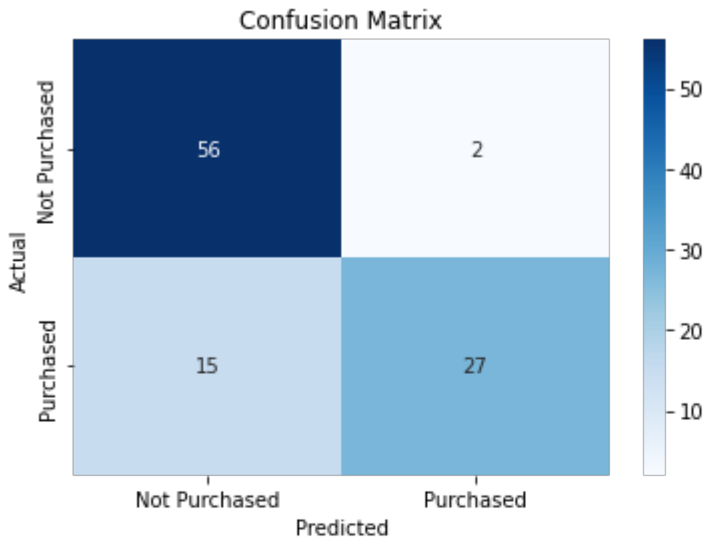
In [7]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score

def compute_metrics(y_true, y_pred):
    cm = confusion_matrix(y_true, y_pred)
    TP = cm[1][1]
    FP = cm[0][1]
    TN = cm[0][1]
    FN = cm[1][0]

    return {"TP": TP, "FP": FP, "TN": TN, "FN": FN}

df = pd.read_csv("Social_Network_Ads.csv")

X = df[['Age', 'EstimatedSalary']]
y = df['Purchased']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.25,
random_state = 42)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

model = LogisticRegression()
model.fit(X_train_scaled, y_train)

y_pred = model.predict(X_test_scaled)
metrics = compute_metrics(y_test, y_pred)

accuracy = accuracy_score(y_test, y_pred)
error_rate = 1 - accuracy
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
```

```
print(f"Confusion Matrix: \n {confusion_matrix(y_test, y_pred)}")
print(f"True Positive (TP): {metrics['TP']}")
print(f"False Positive (TP): {metrics['FP']}")
print(f"True Negative (TN): {metrics['TN']}")
print(f"False Negative (FN): {metrics['FN']}")
print(f"Accuracy: {accuracy: .2f}")
print(f"Error_Rate: {error_rate: .2f}")
print(f"Precision: {precision: .2f}")
print(f"Recall: {recall: .2f}")

sns.heatmap(confusion_matrix(y_test, y_pred), annot = True,
fmt = "d", cmap = "Blues", xticklabels = ["Not Purchased", "Purchased"],
yticklabels=["Not Purchased", "Purchased"])

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```

```
Confusion Matrix:
 [[56  2]
 [15 27]]
True Positive (TP): 27
False Positive (TP): 2
True Negative (TN): 2
False Negative (FN): 15
Accuracy:  0.83
Error_Rate:  0.17
Precision:  0.93
Recall:  0.64
```


Confusion Matrix
Actual
Not Purchased
Purchased
56
2
15
27
Not Purchased
Purchased
Predicted
50
40
30
20
10

In [ ]: